

Dietary animal fat disrupts gut microbiota and aggravates Scl-cGVHD after allogeneic hematopoietic stem cell transfer

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Key Points

- Short-term feeding of an animal-derived fat diet (moderate fat percentage) resulted in worse Scl-cGVHD vs a similar plant-derived fat diet.
- Moderate fat diets resulted in distinct, contracted gut microbiome profiles, which might explain the differences in outcomes.

Allogeneic hematopoietic stem cell transplant (allo-HSCT) is an effective treatment for high-risk or relapsed acute leukemia. However, the frequent occurrence of graft-versus-host disease (GVHD) poses significant complications. Modifiable factors such as the gut microbiome and dietary regimen have the potential to influence the frequency and severity of GVHD. Previous studies in mouse models have shown a direct link between obesity and increased severity of GVHD; however, analysis of human data has not suggested a causal relationship. We hypothesized that dietary fat content before transplantation, rather than obesity itself, might affect outcomes, given that increased dietary fat is associated with reduced gut microbiome diversity. In our study, we evaluated the effects of an animal fat-based diet (AFD) and a plant fat-based diet (PFD) in a mouse model of sclerodermatous chronic GVHD (Scl-cGVHD). Mice initially fed normal chow (~10% kcal from fat) were switched to either an AFD or a PFD (each with ~25% kcal from fat) 2 weeks before allo-HSCT. Mice on the AFD had an elevated serum cytokine response, more severe skin inflammation, and greater intestinal immune dysregulation than mice on the PFD. Although both diets reduced gut microbial diversity, the fat source led to distinct microbiota compositions that included 2 coabundance guilds that differentiated between the AFD and PFD groups. Our findings demonstrate that even short-term consumption of diets with equivalent macronutrient content, but different fat sources, can modulate the gut microbiome, disrupt intestinal homeostasis, and influence Scl-cGVHD outcomes.

Introduction

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is an important treatment for leukemia and related disorders, bone marrow failure syndromes, hemoglobinopathies, immunodeficiency states, and others.¹ Of patients who undergo an allo-HSCT, 30% to 60% will develop acute

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Gut microbiota sequencing data generated in this study have been submitted to the Sequence Read Archive maintained by National Center for Biotechnology Information under the accession number PRJNA1078507.

All data needed to evaluate the conclusions in the paper are present in the paper or the supplemental Materials.

The full-text version of this article contains a data supplement.

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graft-versus-host disease (aGVHD)²⁻⁴ or chronic GVHD (cGVHD).^{5,6} aGVHD is characterized by inflammation and tissue damage of the gastrointestinal tract, liver, and skin.⁷ cGVHD, in contrast, has delayed onset and affects multiple organs and is often dominated by fibrosis.⁸ Although aGVHD and cGVHD have distinct pathologies, the mechanisms driving inflammation have commonalities, including physiological variables, such as the gut microbiome and diet, that can affect allo-HSCT outcomes.⁹⁻¹¹ Loss of gut microbiome diversity as a consequence of antibiotic use, for example, causes a sharp reduction of specific bacterial families that has been correlated with severe GVHD.¹²⁻¹⁸ In patients with GVHD, the loss of microbiota diversity is linked to a reduction of certain short-chain fatty acids (SCFAs) such as butyrate, propionate, and acetate.^{19,20} SCFAs induce immunomodulatory functions that promote intestinal homeostasis. Butyrate, for example, drives extrathymic regulatory T cell (Treg) differentiation,^{21,22} functions as a fuel source for colonocytes, and maintains intestinal barrier function by promoting mucus production and the expression of tight junction proteins.²³⁻²⁵

Restoration of microbiome diversity with third-party fecal matter transplants can increase microbiome diversity in patients after allo-HSCT.^{26,27} Fecal matter transplant has shown promise as a treatment for intestinal GVHD in small clinical studies, however, efficacy has not yet been demonstrated in large controlled studies.²⁸ Prebiotics such as galactooligosaccharide, increase microbial SCFA production and attenuated aGVHD in mouse studies.²⁹⁻³² Various dietary components have been shown to differentially affect gut microbiome composition and GVHD through microbial-dependent and independent mechanisms.¹¹ The level of dietary fat intake is a particularly relevant concern for patients with GVHD, because obesity is highly prevalent in the adult population (~30%)³³ and hyperlipidemia is observed after allo-HSCT.³⁴ Obesity also correlates with decreased gut microbiota diversity and GVHD-associated bacterial taxa.^{15,35,36} Recent mouse studies demonstrated that diet-induced obesity caused by long-term feeding of high-fat diet (HFD) induced gut microbiome dysbiosis and exacerbated disease in both aGVHD and cGVHD mouse models.^{35,36}

We postulated that rather than just total fat content, different sources of dietary fat could divergently affect gut microbiota and, ultimately, cGVHD outcomes in mice. Additionally, although previous studies are informative, the excessively high levels of fat and the use of an aggressive aGVHD model raise concerns about applicability of these results in the context of clinical settings. We therefore established more physiologically relevant model systems that used moderate increases to fat content. Secondly, we used a minor antigen mismatch mouse model (B10.D2→BALB/c), which has pathology similar to human cGVHD.³⁷⁻⁴⁰ Additionally, to analyze the impact of fat sources on cGVHD, we formulated 2 moderate-fat diets that differ only in commonly consumed animal- and plant-derived fat sources. We showed that with just a 2-week exposure to these diets, alterations to the overall gut microbiota structure were evident. Using a model of sclerodermatous cGVHD (Scl-cGVHD), we found that the mice fed an animal-derived fat diet (AFD) had more severe skin and gastrointestinal inflammation, different cytokine and T-cell responses, and succumbed to Scl-cGVHD more quickly than mice fed a plant-derived fat diet (PFD). Collectively, our data show that short-term feeding of diets that are different in fat source affected Scl-cGVHD outcomes in mice.

Materials and methods

Administration of diets

Purchased mice were acclimated and fed normal chow (NC) for at least 2 weeks before initiating experiments. Ghee- or soybean oil-supplemented diets (see supplemental Figure 1A for diet details) were introduced 2 weeks before bone marrow transplantation (BMT). Experimental diets were continued, ad libitum, after BMT.

BMT

BALB/c female mice, aged 8 weeks (The Jackson Laboratory), were fed an AFD or a PFD for 2 weeks before BMT, and remained on their respective diets throughout the study. Irradiated (775 cGy for the cGVHD model, or 850 cGy for the aGVHD model) BALB/c were IV injected with 10×10^6 T-cell thymocyte differentiation antigen 1-depleted bone marrow and 5×10^6 splenic CD5-enriched T cells (Miltenyi beads) from B10.D2 mice. Mice were housed in sterile conditions with hyperchlorinated drinking water. Studies were approved by the Rutgers University institutional animal care and use committee.

GVHD score

Scoring was adapted from Khuat et al³⁵ with the added parameter of level of activity. Clinical scores of weight loss, hunching, decreased activity, diarrhea, fur texture, and skin integrity (0-2 for each), were applied. Mice were euthanized with total score of 9 or 12 or loss of >30% bodyweight. Scl-cGVHD score used a skin scoring system that has been described previously.³⁸

Histology

Samples were prepared as in Bialkowska et al.⁴¹ Hematoxylin and eosin (H&E) and Masson trichrome staining of the skin was done by Nationwide Histology. Inflammation and severity were scored by a board-certified veterinary pathologist (Research Pathology Services, Rutgers University). Colons were scored for inflammation, proprial thickening, and mucosal hyperplasia; skin samples were scored for inflammation and fibrosis. Controls established the threshold for background.

Flow cytometry (FC)

Samples were collected on a Cytex Aurora 4-laser system and analyzed with FlowJo software (Tree Star, Ashland, OR). Nonspecific binding was prevented by blocking with 2% mouse serum, 0.1% anti-Fc γ antibody, and streptavidin. Fixation was done with the FoxP3 buffer set (eBioscience) or the Cyto kit (BD Biosciences). Antibodies used were anti-CD4 (GK1.5), anti-CD62L (MEL-14), anti-CD11b (M1/70), anti-CD19 (1D3), anti-tumor necrosis factor α (TNF- α ; MP6-XT22), anti-CD45 (30-F11), anti-CD3 (500.A2), anti-CD44 (IM7), anti-CD45 (30-F11), anti-CD3 (500.A2), anti-CD8 (53-6.7), anti-FoxP3 (FJK-16s), anti-CD80 (16-10A), anti-F4/80 (BM8), and anti-major histocompatibility complex 1 (212.A1). Dead cells were excluded with Zombie NIR (BioLegend) or Bioscience fixable viability dye eFluor 506 (eBioscience). Doublets were excluded by comparing forward scatter area to height.

Gut permeability

Mice, fasted for 2 hours, were gavaged with fluorescein isothiocyanate (FITC)-dextran (80 mg/mL; Sigma-Aldrich, 46944). Serum

was collected after 5 hours. FITC-dextran was detected at 495 nm excitation and 525 nm emission.

Microbial DNA extraction and 16S rRNA gene V4 region sequencing

Feces samples were collected before, and 2 weeks after, dietary intervention. Microbial genomic DNA was extracted using protocol Q with minor modifications.⁴² The hypervariable region V4 of 16S rRNA gene was amplified using the barcode-tagged 515F⁴³ and 806R⁴⁴ primers. Polymerase chain reaction (PCR) reactions were done with 20 ng of DNA template and the Platinum SuperFi II PCR master mix (Thermo Fisher Scientific). Genomic DNA from a microbial mock community (BEI Resources, Manassas, VA) was used to test consistency between sequencing runs, with no template controls. PCR products were detected on a 1.2% agarose gel electrophoresis, quantified using Qubit 4 (Thermo Fisher Scientific) and purified using AMPure XP (Beckman Coulter, Indianapolis, IN). Amplicons were pooled and then used for preparing

the sequencing chip with the Ion Torrent Chef platform (Thermo Fisher Scientific). The template chip was sequenced using the Ion GeneStudio S5 platform (Thermo Fisher Scientific).

Gut microbiota data analysis

16S rRNA V4 region sequencing was analyzed using the QIIME 2 platform (version 2022.11).⁴⁵ Primers were trimmed from raw reads using cutadapt and an amplicon sequence variants (ASVs) frequency data table was obtained by denoising using the DADA2 pipeline.^{46,47} Spurious ASVs were removed by abundance filtering⁴⁸ and a phylogenetic tree was constructed using the QIIME 2 platform. α - and β -diversity metrics and principal coordinate analysis (PCoA) were calculated by q2-diversity after rarefying the samples to 26 000 sequences per sample. Microbiota structure was compared using β -diversity with Bray-Curtis index between the groups, visualized by PCoA plots and assessed for statistical significance by permutational multivariate analysis of variance (PERMANOVA) via q2-diversity with 9999 permutations.

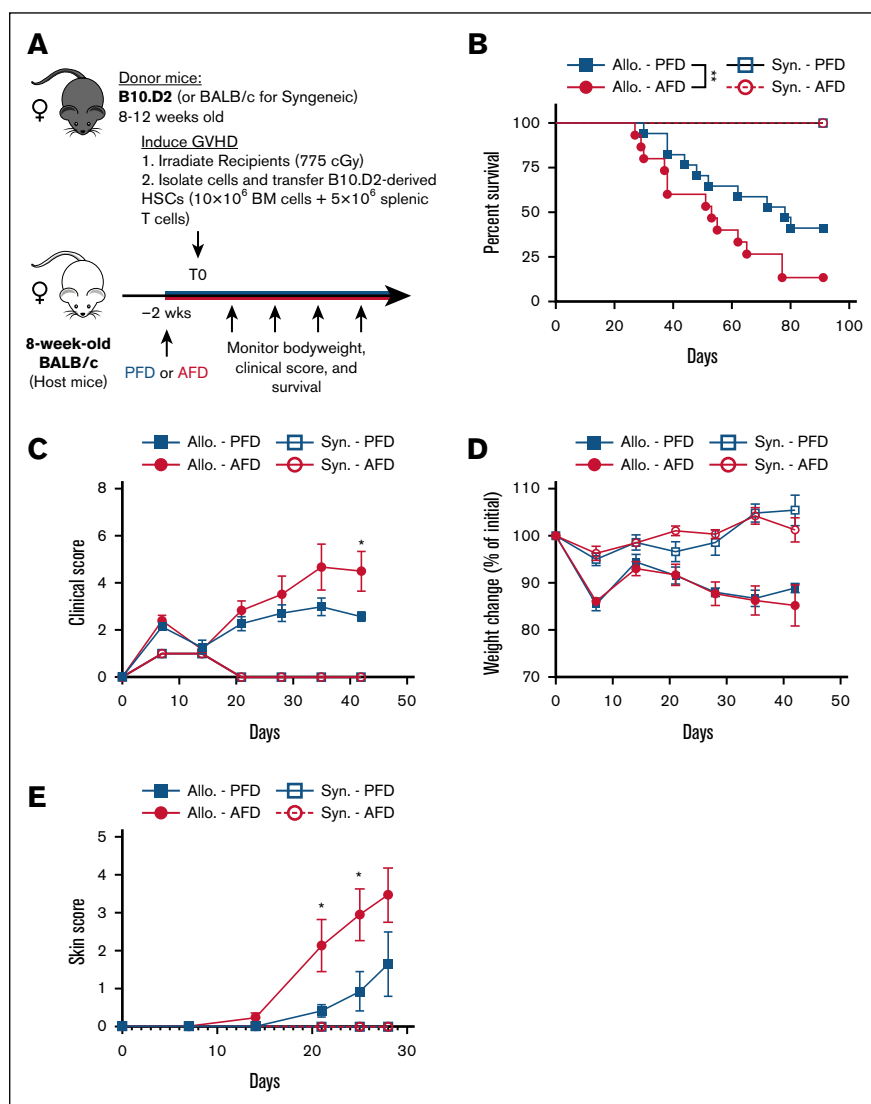


Figure 1. Mice fed AFD have more severe GVHD and worse survival in a model of Scl-cGVHD. Female BALB/c mice (H-2^d) were fed either an AFD or a PFD for 2 weeks before BMT. (A) Recipient mice received total body irradiation of 775 cGy and underwent transplantation with 10×10^6 bone marrow cells and 5×10^6 splenic T cells from either BALB/c (H-2^d, syngeneic graft) or B10.D2 (H-2^d, allogeneic graft) donor mice. (B) Survival rates of all groups shown as Kaplan-Meier curves ($n = 6$ for syngeneic groups, and $n = 15$ for allogeneic groups; pooled cohorts of mice). (C) GVHD clinical scores and (D) body mass changes were measured weekly ($n = 3$ for syngeneic groups, and $n = 7$ for allogeneic groups; data are representative of 2 independent experiments, with 1 cohort shown). (E) Scl-GVHD skin score ($n = 3$ for syngeneic groups, and $n = 6$ for allogeneic groups). Each experiment was performed at least twice. In panel B, Kaplan-Meier survival curve was analyzed by a log-rank test. In panels C-E clinical scores, body mass changes, and skin scores were analyzed by Student *t* test for comparison among allogeneic groups. Survival curves were analyzed by log-rank Mantel Cox test. * $P < .05$, ** $P < .01$. Syn., syngeneic.

Taxonomic assignment was performed by the q2-feature-classifier plugin in QIIME 2 using the SILVA database (release 138).⁴⁹

Guild-based analysis identified coabundance groups (CAGs) while reducing dimensionality of sequencing data.⁵⁰ Correlation between the ASVs shared among >30% of samples were calculated⁵¹ and transformed into a distance matrix (1 – correlation coefficient) for the 170 shared ASVs, which were then grouped into 19 CAGs by applying Ward hierarchical clustering and PERMANOVA with 999 permutations. A cluster tree was built, and PERMANOVA was used in a top-down manner along the tree to identify the nodes that were not significantly different, grouping them into a single CAG ($P > .001$). Procrustes analysis and the Mantel test were performed to assess concordance between the CAG-level data and ASV-level data. Differentially abundant CAGs were identified using a random forest-based Boruta algorithm.⁵² The CAG network for the discriminating CAGs was visualized in Cytoscape version 3.10.1.⁵³ Statistical analysis was conducted on GraphPad Prism. The R package ggplot2 was used for plotting the PCoA plots. Differential abundance of each CAG tested by

Wilcoxon matched-pair signed-rank tests were used to analyze pairwise (ie, NC vs 2 weeks) comparison within each group and a Mann-Whitney test was used to analyze differences between the AFD and PFD groups at the same time point.

Results

Animal fat-supplemented chow exacerbated Scl-cGVHD

Retrospective studies of outcomes after allo-HSCT typically do not show statistically significant changes in the incidence of GVHD in patients with obesity.⁵⁴⁻⁵⁶ These findings are inconsistent with mouse work, which has shown that obesity often strongly correlated with poorer outcomes.^{35,36} We considered the possibility that these discordant results might be, in part, a consequence of dietary factors, such as the source of fat. The fat in food eaten by people is sourced from multiple food products, whereas mouse HFDs are commonly rendered fatty tissue from pigs called lard. Additionally, although fat content in the typical American diet is

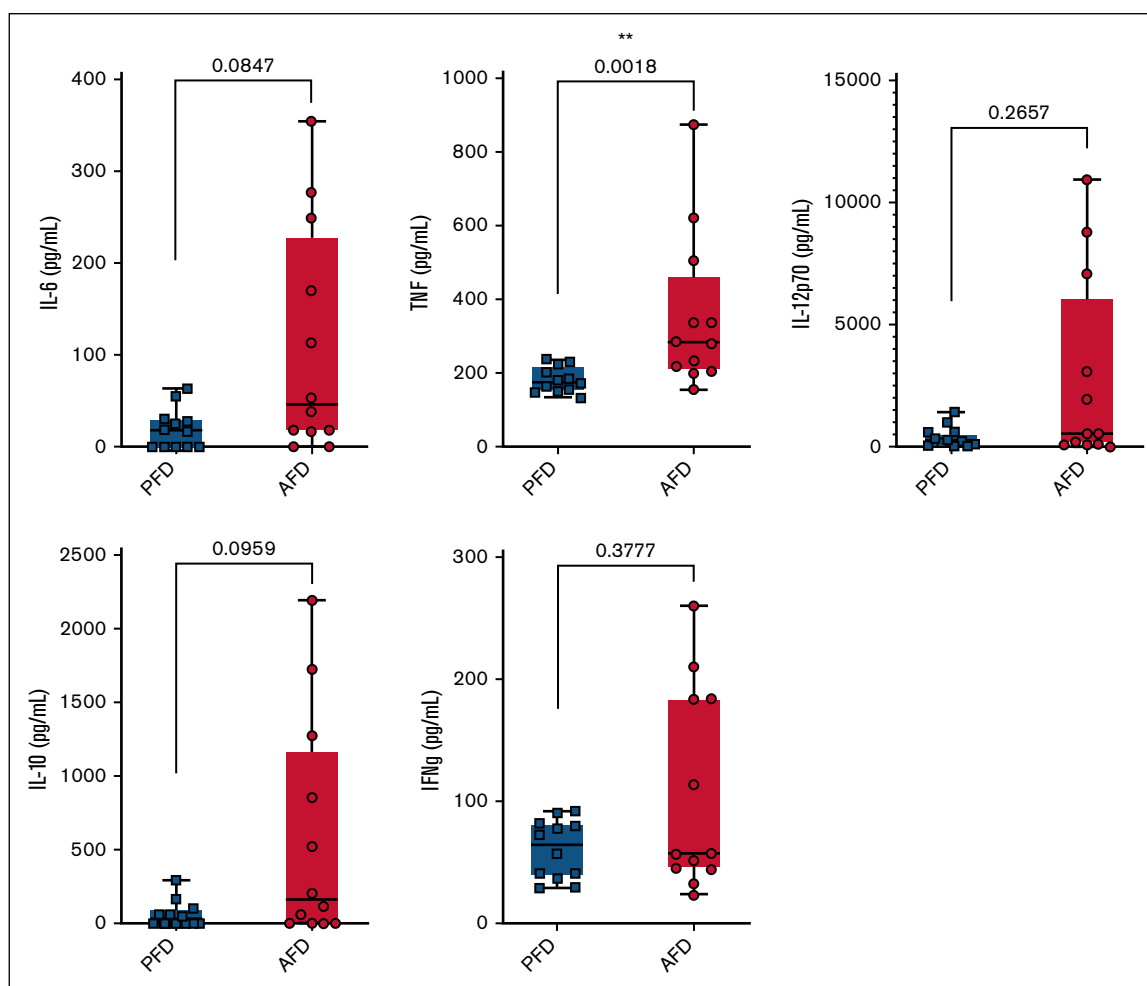


Figure 2. AFD-fed mice have elevated cytokine response during Scl-cGVHD. Comparison of serum cytokine levels in PFD- and AFD-fed mice 14 days after allo-BMT. IL-6, TNF- α , IL-12p70, IL-10, and IFN- γ were measured by cytometric bead array. Total $n = 12$ mice per group. Box plots depict mean and range of data (minimum to maximum). Results were determined to not be normal by use of the Shapiro-Wilk test and, therefore, were analyzed by Mann-Whitney test for comparison between groups. * $P < .05$. INFg, INF- γ .

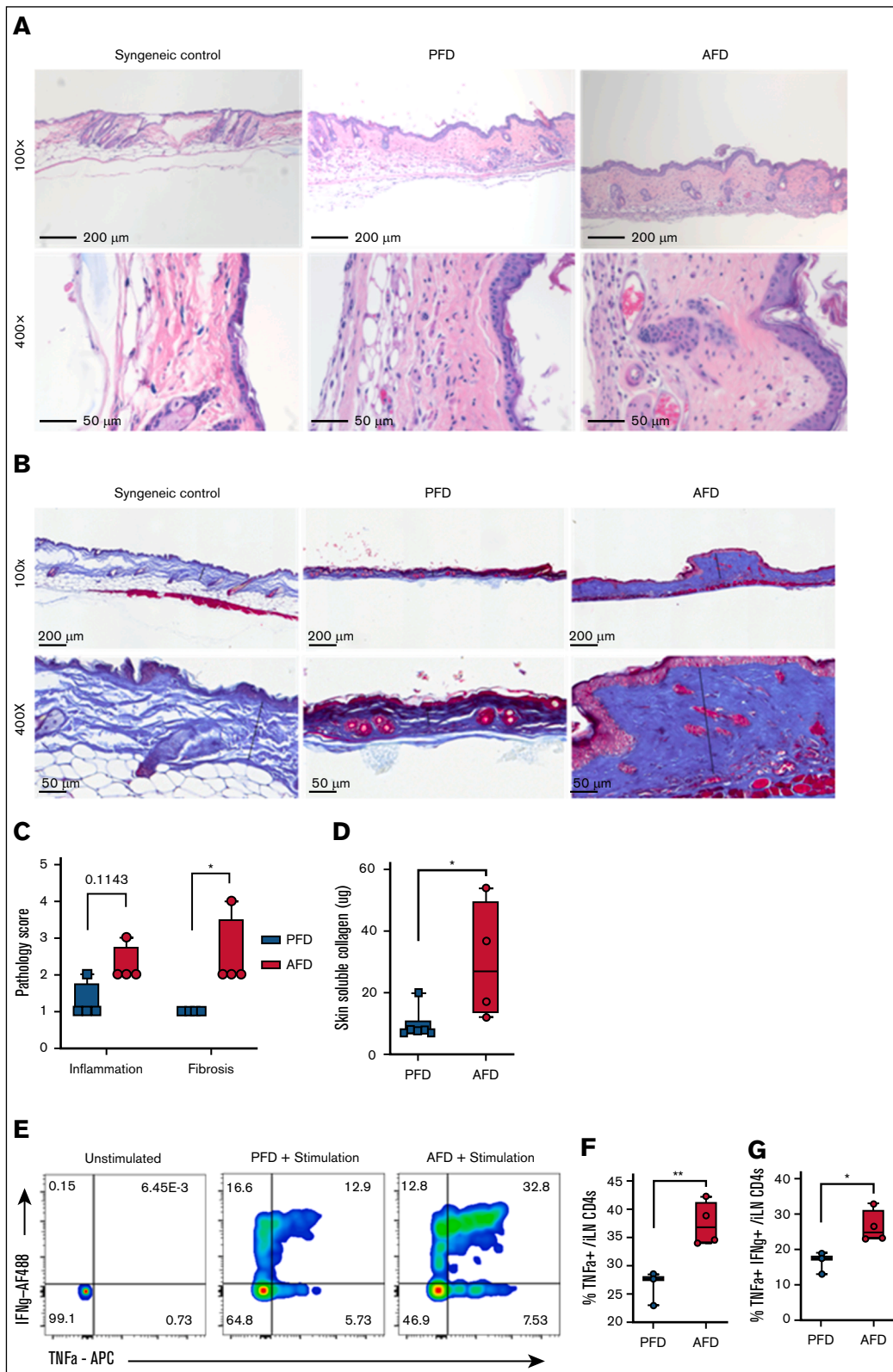


Figure 3. Mice fed an AFD have more severe skin inflammation and fibrosis in Scl-cGVHD. (A) Representative H&E-stained sections of intrascapular skin from AFD-fed syngeneic control vs PFD- and AFD-fed allo-BMT recipients (scale bars 200 μ m for $\times 100$ original magnification, and 50 μ m for $\times 400$ original magnification). (B) Representative Masson trichrome-stained sections of intrascapular skin from AFD-fed syngeneic control vs PFD- and AFD-fed allo-BMT recipients. Black line represents dermal thickness. (C) Comparison of pathology scores depicting degree of skin inflammation and fibrosis between PFD- and AFD-fed allo-BMT recipients. (D) Comparison of concentration of soluble

estimated to be 31% to 38% of total caloric intake,^{33,57} experimental animal diets typically have 60% kcal from fat. We formulated diets with common fat sources, which were the clarified butter, ghee, and soybean oil (full composition of diets is described in supplemental Figure 1A). The fat content of the diets was increased to 25% kcal from fat as compared with ~10% kcal fat in NC.⁵⁸ Therefore, these diets more closely reflected typical human diets. Short-term feeding with these moderate-fat diets did not cause weight gain (supplemental Figure 1B).

To examine the impact of dietary fat source on GVHD, grafts (5×10^6 splenic T cells and 10×10^6 bone marrow cells) were transplanted from B10.D2 (H-2d) donors into lethally irradiated BALB/c mice (H-2d). This minor antigen-dependent disease is less severe than major histocompatibility complex-disparate mouse models of GVHD, which cause death within 2 to 3 weeks.⁵⁹ This model results in the delayed development of fibrotic inflammation in the skin that closely recapitulates human Scl-cGVHD.^{37,39,60} BALB/c host mice were fed ghee-supplemented diet (AFD) or soybean oil-supplemented diet (PFD) starting 2 weeks before BMT (Figure 1A). Diets were continued throughout the course of the study. Mice fed an AFD or a PFD had severe Scl-cGVHD that was lethal in some mice starting ~30 days after the allo-BMT. Interestingly, the survival of AFD-fed allo-BMT recipients was significantly worse compared with PFD-fed mice, with a median survival of 53 vs 78 days, respectively (Figure 1B). Clinical symptoms of GVHD,³⁵ including loss of weight, hunching, decreased activity, diarrhea, change in fur texture, and loss of skin integrity, were more severe in AFD-fed mice (Figure 1C). Both AFD- and PFD-fed allo-BMT recipients had a similar degree of weight loss after HSCT, whereas all syngeneic-BMT control mice fully recovered to their initial bodyweight (Figure 1D). The severity of skin disease was quantified by use of a cumulative score that was based on the area of skin affected by alopecia and the presence of lesions on the ears, paws, and tail, as described previously.³⁸ Data showed that AFD-fed mice developed more severe skin disease within 21 days (Figure 1E). AFD-fed mice, therefore, had more profound skin inflammation, more severe disease, and died sooner than mice fed a PFD after allo-BMT. We also examined the impact of AFD and PFD in a major antigen mismatch model of GVHD in which we transplanted cells from C57BL/6 (H-2b) donor mice into BALB/c (H-2d) recipients. Results showed that fat source did not affect outcomes in this acute model of GVHD (supplemental Figure 1C-E).

Cytokine responses and skin fibrosis are increased in mice fed a AFD

Mouse Scl-cGVHD is driven by a mixed T helper 1/2 cell-like cytokine response in the skin during early fibrosis and requires interleukin-6 (IL-6) signaling to progress.^{38,40} Serum cytokines were measured 14 days after the allo-BMT, which preceded overt tissue damage and loss of skin integrity. The complexity of these

outcomes and the variability on the time of onset was reflected by the data at this single time point. There were trends for increased levels of multiple cytokines, including IL-6, TNF, IL-12p70, IL-10, and interferon gamma (IFN- γ), however, statistical analysis of these nonlinear data sets showed that only the change to TNF- α reached significance (Figure 2).

Histopathological scores and collagen deposition were measured by staining skin sections with H&E and Masson trichrome stains, respectively (Figure 3A-B). Mice fed the AFD had significantly more fibrosis than mice fed the PFD (Figure 3C). Recently synthesized collagen (immature biochemical collagen) can be solubilized by an acid-pepsin digestion, whereas mature structural collagen is resistant to acid-pepsin digestion. To measure recently synthesized collagen, intrascapular skin samples (~6 mm in size) harvested at end point were homogenized in an acid pepsin solution for 48 hours and after centrifugation, and the supernatant fraction was assessed with a collagen-specific dye. On average, AFD-fed recipients had greater than a threefold increase in the amount of soluble collagen in the skin compared with PFD-fed mice (Figure 3D), consistent with more severe skin disease.

Finally, T cells in skin-draining inguinal lymph nodes were stimulated with phorbol 12-myristate 13-acetate and ionomycin and analyzed by FC for TNF- α , IFN- γ , IL-4, and IL-17A (Figure 3E; supplemental Figure 2A). AFD-fed mice had a significantly higher frequency of TNF- α^+ CD4 T cells (Figure 3F) and a higher frequency of TNF- α^+ IFN- γ^+ CD4 T cells than PFD-fed mice (Figure 3G). The elevated TNF- α response is of particular interest because it can both initiate and accelerate GVHD pathology.⁶¹ IL-4- and IL-17A-producing CD4 T cells were nearly undetectable (supplemental Figure 2B). Overall, these data indicated that mice fed a AFD had more profound Scl-cGVHD and skin inflammation than mice fed a PFD, and that this was associated with increased TNF- α production by CD4 T cells in the skin.

AFD-fed mice have enhanced immune dysregulation in the gut during Scl-cGVHD

Damage to the intestine was tested by oral gavage of fasted mice with FITC-dextran.⁶² AFD-fed allo-BMT recipients had significantly more FITC-dextran in the serum as compared with PFD-fed recipients, indicating a loss of barrier integrity in AFD-fed recipients (Figure 4A). Lipocalin-2 is upregulated during inflammation and is released by epithelial cells and macrophages as a consequence of inflammation.⁶³⁻⁶⁶ Consistent with worse disease, AFD-fed mice had ~approximately fivefold higher levels of lipocalin-2 than PFD-fed mice 28 days after allo-BMT (Figure 4B).

Cells isolated from the colons and mesenteric lymph nodes (mLNs) of PFD- and AFD-fed allo-BMT recipients at end point were analyzed by FC (gating scheme outlined in supplemental Figure 2A). AFD-fed mice had approximately half as many Tregs in both the colonic lamina propria and the epithelium as compared

Figure 3 (continued) collagen in the intrascapular skin of PFD- and AFD-fed allo-BMT recipients. (E) Representative FC plots depicting IFN- γ and TNF- α cytokine production by CD4 $^+$ T cells isolated from the skin-draining inguinal lymph node of PFD- or AFD-fed allo-BMT recipients at end point. (F) Results quantified as percentage of total TNF- α -producing CD4 $^+$ T cells and (G) TNF- α and IFN- γ double-producing CD4 $^+$ T cells. For panels A-D, total $n = 4$ mice per group; for panels E-G, analysis was performed on mice as they reached euthanasia criteria between days 35 and 49 after transplant. Data points show independently collected samples from 7 mice (4 AFD, 3 PFD). Bar graphs depict mean $\pm$ standard error of the mean (SEM). Results were analyzed by Student's t test for comparison between groups, except when data sets did not pass the Shapiro-Wilk test for normality, then data were analyzed by Mann-Whitney test for comparison between groups (panels C-D). * $P < .05$, ** $P < .01$. INFg, INF- γ ; iLN, inguinal lymph node.

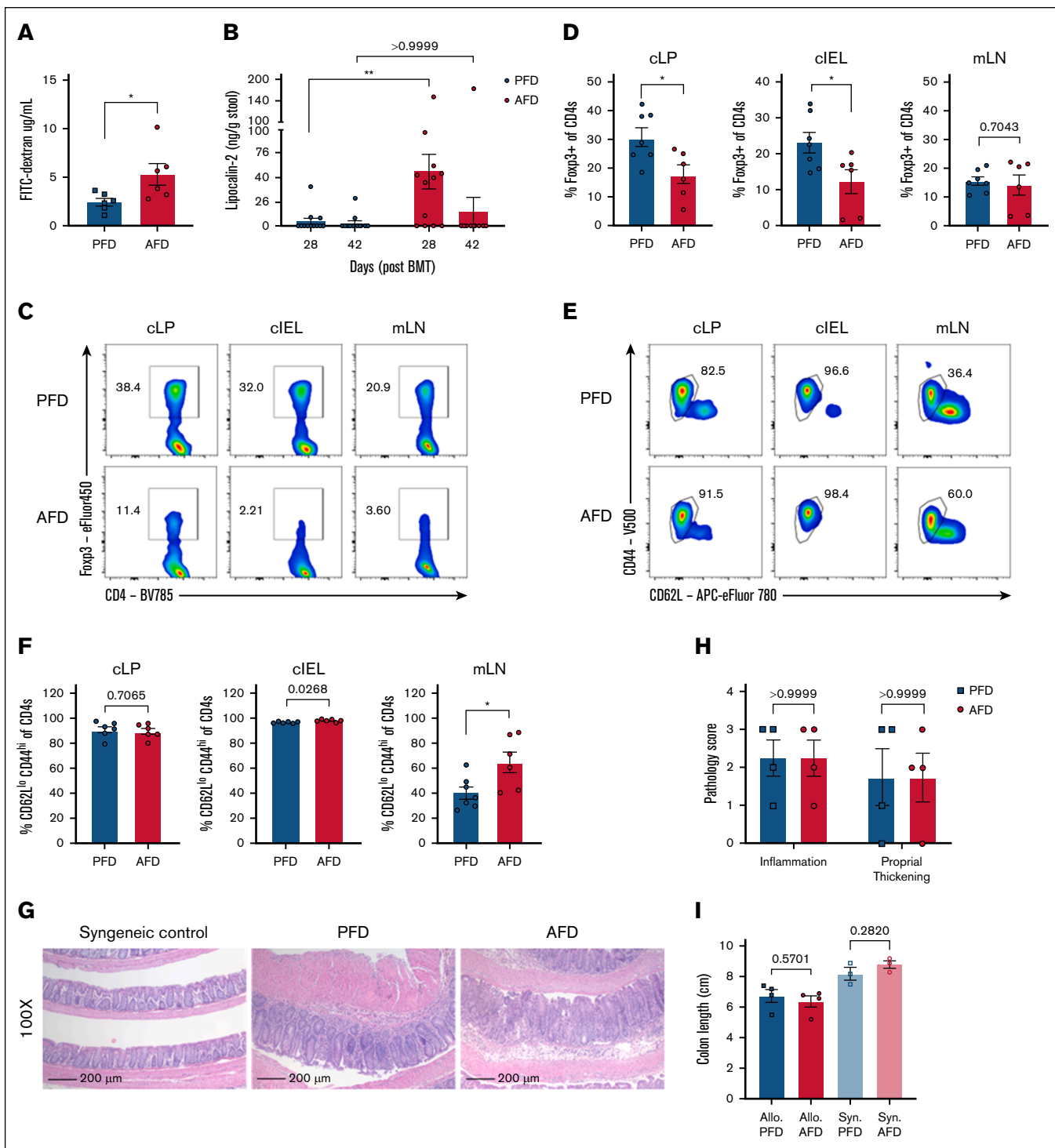


Figure 4. AFD-fed mice have enhanced immune dysregulation in the gut during Scl-cGVHD. (A) FITC-dextran translocation across the gastrointestinal barrier into blood serum in PFD-fed allo-BMT recipients compared with AFD-fed allo-BMT recipients 21 days after BMT. (B) Comparison of fecal lipocalin-2 levels in PFD- and AFD-fed mice 28 days after allo-BMT. (C) Representative FC plots of CD4⁺ Foxp3⁺ Tregs in the mLN cIEL and cLP of PFD- and AFD-fed allo-BMT recipients and (D) quantification as percentage of total CD4⁺ cells. (E) Representative FC plots of CD62L and CD44 expression in CD4⁺ T cells from mLN cIEL and cLP of PFD- and AFD-fed allo-BMT recipients. (F) Quantification of activated CD62L^{lo}CD44^{hi}CD4⁺ T cells as percentage of total CD4⁺ T cells. (G) Representative H&E-stained sections of the colons from PFD- and AFD-fed allo-BMT recipients (scale bars, 200 μ m). (H) Comparison of colon pathology scores for inflammation and proprial thickening between PFD- and AFD-fed allo-BMT recipients. (I) Colon lengths of PFD- and AFD-fed allo-BMT recipients. All data were collected at end point (35-49 days after BMT) unless otherwise specified. For panels A, C-E, total n = 6 mice per group; for panel B, total n = 12 mice per group; and for panels G-I, total n = 4 mice per group. Bar graphs depict mean $\pm$ SEM. Results were analyzed by Student *t* test for comparison between groups, except when data sets did not pass the Shapiro-Wilk test for normality, then data were analyzed by Mann-Whitney test for comparison between groups (panels B, H). **P* < .05, ***P* < .01. cIEL, colonic intraepithelial lymphocytes; cLP, colonic lamina propria; Syn., syngenic.

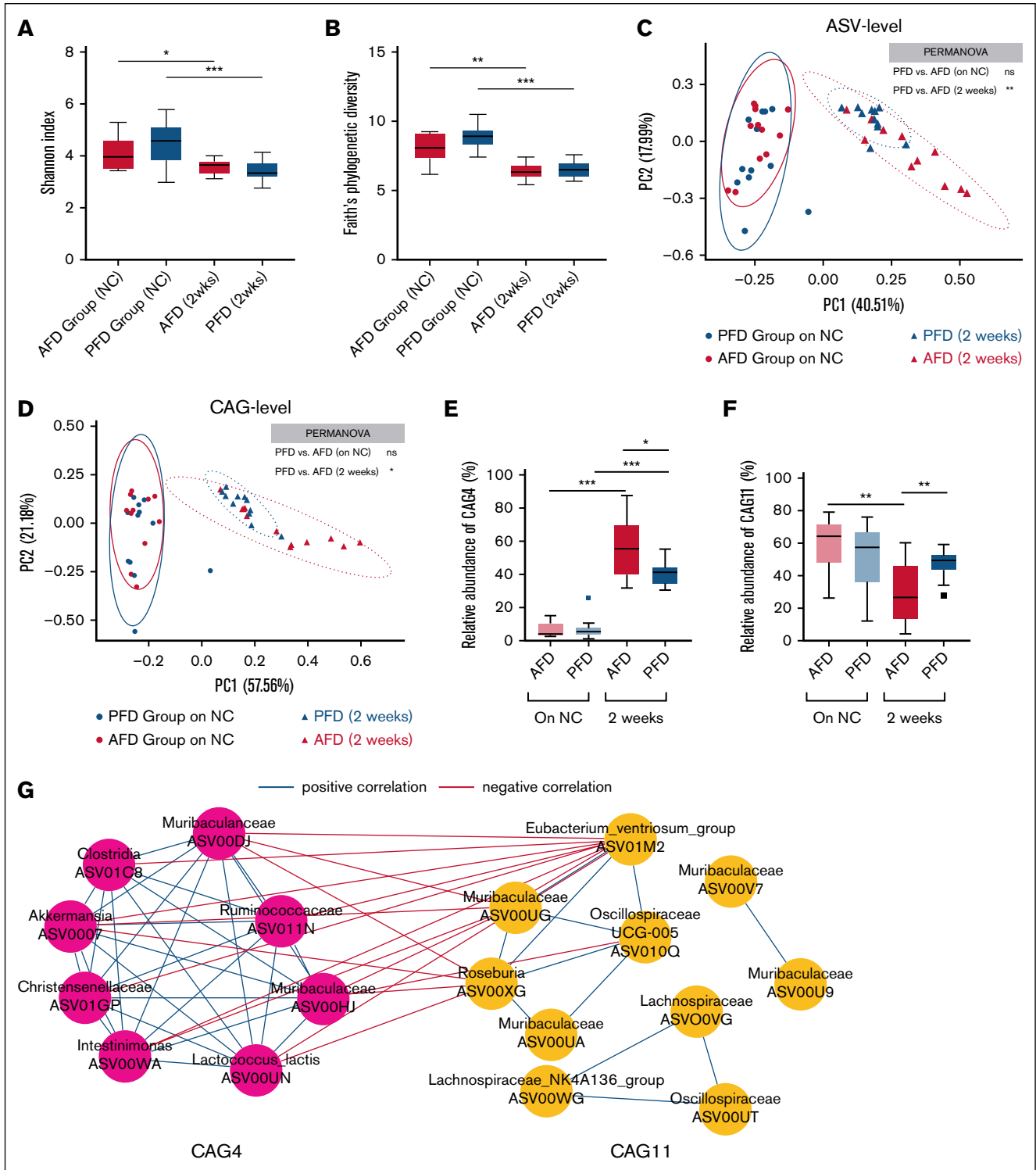


Figure 5. AFD induces a unique gut microbiome in mice. Stool was collected from all mice before dietary intervention (at baseline; on NC) and after 2 weeks of dietary intervention (steady state mice fed the AFD or PFD, before GVHD induction) and analyzed for gut microbiome bacterial strains using 16S rRNA gene sequencing. After stool collection, GVHD was induced, and mice were followed up for survival analysis. From analysis of the 16S rRNA gene sequencing data, α -diversity was estimated as measures of (A) Shannon index and (B) Faith phylogenetic diversity. The box plots show the medians and the interquartile ranges (IQRs), with the whiskers indicating the lowest and highest values that were 1.5 times the IQR from the first and third quartiles. A Mann-Whitney test was used to compare the 2 treatment groups (PFD vs AFD). A paired Wilcoxon test was used to compare the 2 time points in each group. (C) PCoA plot of the gut microbiota structure based on the Bray-Curtis distance with corresponding PERMANOVA comparisons (9999 permutations) for the AFD or PFD groups vs their respective microbiomes at baseline (ellipses represent 95% confidence interval [CI]) at the ASV level.

with PFD-fed mice (Figure 4C-D). Although similar frequencies of Tregs were found in the mLNs, the CD4 T cells from AFD-fed mice were more activated as shown by CD62L^{lo}CD44^{hi} expression (Figure 4F). Virtually all CD4 T cells in the colonic lamina propria and colonic epithelium were CD62L^{lo}CD44^{hi} expression across both groups of mice (Figure 4E-F). mLN CD4 T cells from the 2 groups of mice did not express different levels of TNF- α , IFN- γ , and IL-17A (supplemental Figure 2C). PFD- and AFD-fed mice had no significant differences in the frequency of colonic macrophages or proinflammatory macrophages (supplemental Figure 3A-B).

Histological evaluation of H&E-stained colon sections showed that allo-BMT recipients (both PFD- and AFD-fed mice) had heightened inflammation and enhanced propria thickening compared with syngeneic-BMT control mice (Figure 4G). However, there were no significant differences between the allogeneic groups (Figure 4H). Consistent with the histological findings, PFD- and AFD-fed allo-BMT recipients both had shorter colons than syngeneic controls, indicating intestinal inflammation, but there was no significant difference between groups (Figure 4I).

AFD induced a unique gut microbiome in mice

Differences in microbiota are linked to GVHD in mice and humans.^{24,67-69} To test the impact of fat sources on the microbiome, fecal pellets were collected from littermate mice that were fed NC and then collected again from the same mice after 2 weeks of feeding with either the AFD or PFD, before BMT. After stool collection, Scl-cGVHD was induced and disease onset was tracked. The hypervariable V4 region of the 16S rRNA gene was sequenced from DNA samples extracted from fecal pellets. Both diets caused a significant reduction of gut microbiome diversity as demonstrated by a reduction in both Shannon index and Faith phylogenetic diversity (Figure 5A-B). However, the structure of the gut microbiotas were significantly different, as illustrated in the PCoA plot using the Bray-Curtis dissimilarity based on all the 467 ASVs and confirmed by PERMANOVA analysis (Figure 5C). As expected, there were no significant differences in microbiota among the mice before the dietary change (Figure 5C). These results were consistent in an independent experiment (supplemental Figure 4). Together, these data showed that 2 weeks of the PFD or AFD significantly affected the gut microbiota structure in distinct ways.

To understand the differences between the microbiome in AFD- vs PFD-fed mice, we used guild-based analysis, which operates on the basis that members of an ecosystem seldom exist in isolation; instead, they develop local interactions and form intermember organizations to influence the ecosystem's higher-level patterns and functions.⁷⁰ Such intermember organizations, or guilds, have members that exhibit coabundant behavior.⁵⁰ Coabundance relationships between the 170 prevalent ASVs that were shared by at least 30% of the samples were determined. These 170 ASVs

accounted for ~92% of the total sequences that formed 19 different CAGs. PCoA and PERMANOVA analysis (Figure 5D) for the 19 CAGs using the Bray-Curtis dissimilarity revealed 2 groups at baseline, which had similar structure and that, after 2 weeks on their respective diets, developed significantly distinct structures.

Procrustes analysis and Mantel test showed a significant correlation between β -diversity results based on the 467 ASVs and all 19 CAGs (supplemental Figure 5A-B). This implies that the CAG-level data provided a faithful representation of the overall gut microbiota structure as described with the ASV-level data. To identify the differentially abundant CAGs, random forest-based Boruta algorithm⁵² was applied. CAG4 and CAG11 were found to discriminate between PFD and AFD mice after 2 weeks of dietary intervention (supplemental Figure 5C) with an area under the receiver operating characteristic curve of 0.78 (supplemental Figure 5D). CAG4 was significantly increased in the AFD group compared with the PFD group (Figure 5E). CAG11 was more abundant in PFD-fed mice by nearly twofold compared with in AFD-fed mice (Figures 5F). Furthermore, the relative abundance of CAG11 decreased significantly in the AFD group compared with its baseline abundance, whereas changes in CAG11 abundance in the PFD group were not statistically significant (Figure 5F). CAG4 had a negative correlation trend with the number of survival days after allo-HSCT whereas CAG11 had a positive correlation trend with survival (supplemental Figure 5E-F); neither correlation, however, reached statistical significance ($P = .1073$ and $P = .1801$, respectively). CAG4 contained 1 ASV from genus *Akkermansia*, 1 ASV from genus *Intestinimonas*, 1 ASV from genus *Lactococcus*, 2 ASVs from family *Muribaculaceae*, 1 ASV from family *Christensenellaceae*, 1 ASV from class *Clostridia*, and 1 ASV from family *Ruminococcaceae* (Figure 5G). Conversely, CAG11, consisted of 10 ASVs, 4 ASVs from family *Muribaculaceae*, 2 ASVs from family *Oscillospiraceae*, 2 ASVs from *Lachnospiraceae*, 1 ASV from genus *Roseburia*, and 1 ASV from genus *Eubacterium ventriosum* group (Figure 5G).

These data show that a moderate increase in fat was sufficient to alter the overall structure of gut microbiota in a diet-specific manner. Two potential guilds responded to the 2 diets in opposite directions, which were associated with differences in inflammatory response and GVHD, in mice fed the AFD vs PFD.

Discussion

We studied the impact that short-term ingestion of diets made with different fat sources had on Scl-cGVHD outcomes in mice receiving allo-HSCT. The diets had either animal fat from the clarified butter, ghee, or plant fat from the seeds of soybean plants, as opposed to rendered fatty tissue from pigs (lard), which is commonly used in

Figure 5 (continued) (D) PCoA plot of the gut microbiota structure based on the Bray-Curtis distance with corresponding PERMANOVA comparisons (9999 permutations) for the AFD or PFD groups vs their respective microbiomes at baseline (ellipses represent 95% CI) at the CAG-level. (E) CAG4 was identified to be more abundant in AFD mice and (F) CAG11 was identified to be more abundant PFD mice. Wilcoxon matched-pair signed-rank tests were used to analyze pairwise (ie, NC vs 2 weeks) comparison within each group and a Mann-Whitney test was used to analyze differences between the AFD and PFD groups at the same time point. (G) Coabundance network of the 2 discriminating CAGs with the individual ASVs, edges between the nodes denote correlation. Red and blue lines indicate negative and positive correlations, respectively. Correlations with an absolute value of >0.4 and adjusted P value of $<.1$ are shown. P values were adjusted by the Benjamini, Krieger, and Yekutieli method. For panels A-C, total $n = 12$ mice per group; for panels E-G, total $n = 9$ mice per group. * $P < .05$, ** $P < .01$, *** $P < .001$.

rodent HFDs. The diets had macronutrient profiles similar to normal rodent diets, but the percentage of kcal derived from fat was elevated from 10% to 25%.^{57,58,71} By use of commonly consumed foods and a percentage of calories from fat similar to that found in typical human diets,⁵⁷ we suggest that results from these studies more closely approximate the potential dietary impact of fat source in the clinical setting of human allo-HSCT.

Induction of Scl-cGVHD in mice fed the AFD resulted in worsened skin disease and other clinical manifestations, which resulted in more deaths compared with recipients fed the PFD. In the early stages of disease, mice fed an AFD had elevated serum cytokine responses, consistent with more severe skin inflammation. The mice fed with AFD also had intestinal disruptions including increased gut permeability and gut innate immune activation as reflected by elevated fecal lipocalin-2. Although signs of immune dysregulation in AFD-fed mice, such as a loss of Tregs in the colon, were detectable at a late stage, AFD- and PFD-fed mice had a similar degree of overall colonic inflammation measured by colon pathology and length.

Studies have shown that decreased microbiome diversity is directly correlated with more severe GVHD, and that microbiome metabolites can improve GVHD.^{14,16,19,24} Additionally, more severe Scl-cGVHD that resulted from experiments in which mice were fed a 60% HFD over 4 to 6 months, were also associated with a marked reduction of gut microbiome diversity.³⁵ Remarkably, our results showed that a brief exposure to diet with moderately increased fat caused a contraction of gut microbial diversity. Moreover, the resultant microbiome was distinct in animal fat-fed vs plant fat-fed mice. Analysis revealed 2 CAGs that discriminated between PFD- and AFD-fed mice. CAG4, which was more highly abundant in AFD-fed mice, correlated with, and possibly caused, the heightened inflammation and more severe Scl-GVHD seen in AFD-fed mice. Notably, CAG4 was found to contain an ASV from the genus *Akkermansia*. Members of this genera have been found to be associated with GVHD^{15,35} and are elevated in other adverse inflammatory contexts such as dextran sulfate sodium-induced colitis.⁷²⁻⁷⁴ Conversely, CAG11 was more highly abundant in PFD-fed mice and correlated with better survival after allo-HSCT, suggesting a potential beneficial role in host health. CAG11 contained several phylotypes that have been found to be protective in mouse models of colitis, including the genus *Roseburia*, family *Oscillospiraceae*, and family *Lachnospiraceae*.⁷⁵⁻⁷⁷ These data support the concept that the AFD disrupted gut microbiota by promoting CAG4 and diminishing CAG11, which may have contributed to the more severe inflammatory response and cGVHD-associated lethality. Combined, these data show that dietary fat source, rather than dietary fat percentage alone, can have a significant impact on gut microbial diversity and Scl-cGVHD severity.

Our data, although limited to 1 cGVHD model, provide support for the premise that dietary intervention could be a means to affect

GVHD. It will be interesting to learn how these findings relate to the clinical setting. For instance, investigating whether dietary patterns associate with patient outcomes and affect gut microbiota structure at the time of transplant may provide valuable insights for developing GVHD prophylactic and, even potentially, treatment strategies. In conclusion, improving our understanding of the complex interplay and immunomodulatory effects of nutritional factors and gut microbiota may provide valuable insights into GVHD pathogenesis and management, and may validate dietary intervention as a safe and accessible adjunct therapy for allo-HSCT recipients.

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Authorship

Contribution: D.D.M., R.S., and D.B.S. conceived the overall project and designed the experimental plan; D.D.M., S.G., and G.W. planned and carried out experiments, with critical input from A.K.K.; L.O. assisted with mouse work; D.D.M., S.G., G.W., L.Z., R.S., and D.B.S. discussed and interpreted data; D.D.M. and D.B.S. wrote the manuscript; and R.S., S.G., G.W., L.Z., and A.K.K. critically reviewed the manuscript.

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